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FASTENING DEVICE

Abstract

A fastening device includes a case including a plurality of engaging teeth, a spool disposed at the case and configured for a lace to be wound therearound, a knob covering the case, and an engaging unit disposed at the case and coupled to the spool. The engaging unit includes at least one pawl arm selectively coupled to at least one of the engaging teeth, and at least one abutting member. As the knob is rotated in a tensioning direction, the engaging unit is rotated and the pawl arm is disengaged from the engaging teeth to allow the spool to tension the lace. As the knob is rotated in a releasing direction to allow the pawl arm to disengage from the engaging teeth, the abutting member is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a Continuation of the U.S. application Ser. No. 18/227,927, filed Jul. 29, 2023, which claims priority to Taiwan Application Serial Number 111128831, filed Aug. 1, 2022, and Taiwan Application Serial Number 112125406, filed Jul. 7, 2023, all of which are herein incorporated by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a fastening device. More particularly, the present disclosure relates to a fastening device for releasing or tensioning a lace.

Description of Related Art

[0003] In daily life, cords, such as a lace or a thread, are usually used to tighten articles. The most common tightening method is to use the cord to reciprocally pass through holes on the article, such as eyelets of a shoe, and then tie a knot to secure the article. But in this kind of tightening method, the knot is loosened easily because of an external force. Not only does the knot need to be tied again, but also lots of inconveniences come owing to the insecurity of the articles.

[0004] In order to solve such problems, some practitioners developed a simple fastening mechanism including a case, a driving unit and a spring. The case includes holes configured for the lace to pass therethrough. Through the reaction force between the spring and the driving unit, the lace can be clamped between the driving unit and the case so as to be fastened. The length of the lace can be changed by pressing the spring to change the position of the driving unit. However, in such fastening mechanism, the restoring force of the spring is served as the securing force; thus, the lace is easily to be released owing to vibrations or an external force. In addition, the fastening mechanism has no space to receive the lace, and the exposure of the lace may bring danger.

[0005] Therefore, some practitioners developed another kind of buckle which may be rotated to tension the cord, the cord may be received inside the buckle, and through the interference between components inside the buckle, the length of the cord as well as the tightness may be adjusted. However, the structure of the buckle is complex, such that the manufacturing cost is increased, and the buckle has assembly and repair difficulties. In addition, as releasing the buckle, only full release function can be practiced, but the cord (lace) cannot be sectionally (incrementally) released, and there is still an inconvenience of usage.

[0006] Based on the aforementioned problems, how to improve the structure of the fastening device such as the buckle to include an incremental release function becomes a target that those in the industry pursue.

SUMMARY

[0007] According to one aspect of the present disclosure, a fastening device includes a case including a plurality of engaging teeth, a spool disposed at the case and configured for a lace to be wound therearound, a knob covering the case, and an engaging unit disposed at the case and coupled to the spool. The engaging unit includes at least one pawl arm selectively coupled to at least one of the engaging teeth, and at least one abutting member. As the knob is rotated in a tensioning direction, the engaging unit is rotated and the at least one pawl arm is disengaged from at least one of the engaging teeth to allow the spool to tension the lace. As the knob is rotated in a

releasing direction to allow the at least one pawl arm to disengage from at least one of the engaging teeth, the at least one abutting member is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.

[0008] According to another aspect of the present disclosure, a fastening device includes a case including a plurality of engaging teeth, a spool disposed at the case and configured for a lace to be wound therearound, a knob covering the case and including a plurality of knob teeth, and an engaging unit disposed at the case and coupled to the spool. The engaging unit includes at least one pawl arm selectively coupled to at least one of the engaging teeth. As the knob is rotated in a tensioning direction, the engaging unit is rotated to drive the spool to tension the lace, and as the knob is rotated in a releasing direction, the at least one pawl arm is incrementally and radially deflected, a tension of the lace causes the at least one pawl arm to be fully disengaged from at least one of the engaging teeth to offset a friction temporarily, thereby allowing the lace to be incrementally released.

[0009] According to yet another aspect of the present disclosure, a fastening device includes a case including a plurality of engaging teeth, a spool disposed at the case and configured for a lace to be wound therearound, a knob covering the case, at least one pawl arm selectively coupled to at least one of the engaging teeth, and at least one abutting member. As the knob is rotated in a tensioning direction, the at least one pawl arm is disengaged from at least one of the engaging teeth to allow the spool to tension the lace, and as the knob is rotated in a releasing direction to allow the at least one pawl arm to disengage from at least one of the engaging teeth, the at least one abutting member is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

[0011] FIG. 1 is a three-dimensional schematic view of a fastening device according to a first embodiment of the present disclosure.

[0012] FIG. 2 is one exploded view of the fastening device of the first embodiment of FIG. 1.

[0013] FIG. 3 is another exploded view of the fastening device of the first embodiment of FIG. 1.

[0014] FIG. 4 is a cross-section top view of the fastening device of the first embodiment of FIG. 1.

[0015] FIG. 5 is an operation shown by partial cross-section top views of the fastening device of the first embodiment of FIG. 1.

[0016] FIG. 6 is one cross-section side view of the fastening device of the first embodiment of FIG. 1.

[0017] FIG. 7 is another cross-section side view of the fastening device of the first embodiment of FIG. 1.

[0018] FIG. 8 is a three-dimensional schematic view of a fastening device according to a second embodiment of the present disclosure.

[0019] FIG. 9 is one exploded view of the fastening device of the second embodiment of FIG. 8.

[0020] FIG. 10 is another exploded view of the fastening device of the second embodiment of FIG. 8.

[0021] FIG. 11 is a cross-section top view of the fastening device of the second embodiment of FIG. 8.

[0022] FIG. 12 is one cross-section side view of the fastening device of the second embodiment of FIG. 8.

[0023] FIG. 13 is another cross-section side view of the fastening device of the second embodiment

of FIG. 8.

[0024] FIG. 14 is one exploded view of a fastening device according to a third embodiment of the present disclosure.

[0025] FIG. 15 is another exploded view of the fastening device of the third embodiment of FIG. 14.

[0026] FIG. 16 is a three-dimensional cross-section view of the fastening device of the third embodiment of FIG. 14.

[0027] FIG. 17 is a cross-section top view of the fastening device of the third embodiment of FIG. 14.

[0028] FIG. 18 is one cross-section side view of the fastening device of the third embodiment of FIG. 14.

[0029] FIG. 19 is another cross-section side view of the fastening device of the third embodiment of FIG. 14.

[0030] FIG. 20 is one exploded view of a fastening device according to a fourth embodiment of the present disclosure.

[0031] FIG. 21 is another exploded view of the fastening device of the fourth embodiment of FIG. 20.

[0032] FIG. 22 is one cross-section top view of the fastening device of the fourth embodiment of FIG. 20.

[0033] FIG. 23 is another cross-section top view of the fastening device of the fourth embodiment of FIG. 20.

[0034] FIG. 24 is one cross-section side view of the fastening device of the fourth embodiment of FIG. 20.

[0035] FIG. 25 is another cross-section side view of the fastening device of the fourth embodiment of FIG. 20.

[0036] FIG. 26 is one exploded view of a fastening device according to a fifth embodiment of the present disclosure.

[0037] FIG. 27 is another exploded view of the fastening device of the fifth embodiment of FIG. 26.

[0038] FIG. 28 is one cross-section top view of the fastening device of the fifth embodiment of FIG. 26.

[0039] FIG. 29 is another cross-section top view of the fastening device of the fifth embodiment of FIG. 26.

[0040] FIG. 30 is one cross-section side view of the fastening device of the fifth embodiment of FIG. 26.

[0041] FIG. 31 is another cross-section side view of the fastening device of the fifth embodiment of FIG. 26.

[0042] FIG. 32 is one exploded view of a fastening device according to a sixth embodiment of the present disclosure.

[0043] FIG. 33 is another exploded view of the fastening device of the sixth embodiment of FIG. 32.

[0044] FIG. 34 is a partial three-dimensional schematic view of the fastening device of the sixth embodiment of FIG. 32.

[0045] FIG. 35 is one cross-section top view of the fastening device of the sixth embodiment of FIG. 32.

[0046] FIG. 36 is another cross-section top view of the fastening device of the sixth embodiment of FIG. 32.

[0047] FIG. 37 is one cross-section side view of the fastening device of the sixth embodiment of FIG. 32.

[0048] FIG. 38 is another cross-section side view of the fastening device of the sixth embodiment

of FIG. 32.

[0049] FIG. 39 is one exploded view of a fastening device according to a seventh embodiment of the present disclosure.

[0050] FIG. 40 is another exploded view of the fastening device of the seventh embodiment of FIG. 39.

[0051] FIG. 41 is one cross-section side view of the fastening device of the seventh embodiment of FIG. 39.

[0052] FIG. 42 is another cross-section side view of the fastening device of the seventh embodiment of FIG. 39.

[0053] FIG. 43 is one exploded view of a fastening device according to an eighth embodiment of the present disclosure.

[0054] FIG. 44 is another exploded view of the fastening device of the eighth embodiment of FIG. 43.

[0055] FIG. 45 is one cross-section side view of the fastening device of the eighth embodiment of FIG. 43.

[0056] FIG. 46 is another cross-section side view of the fastening device of the eighth embodiment of FIG. 43.

DETAILED DESCRIPTION

[0057] It will be understood that when an element (or mechanism or module) is referred to as being “disposed on”, “connected to” or “coupled to” another element, it can be directly disposed on, connected or coupled to the other element, or intervening elements may also be present. In contrast, when an element is referred to as being “directly disposed on”, “directly connected to” or “directly coupled to” another element, there are no intervening elements present. In addition, the terms first, second, third, etc. are used herein to describe various elements or components, these elements or components should not be limited by these terms. Consequently, a first element or component discussed below could be termed a second element or component.

[0058] FIG. 1 is a three-dimensional schematic view of a fastening device 1000 according to a first embodiment of the present disclosure. FIG. 2 is one exploded view of the fastening device 1000 of the first embodiment of FIG. 1. FIG. 3 is another exploded view of the fastening device 1000 of the first embodiment of FIG. 1. The fastening device 1000 includes a case 1100 including a plurality of engaging teeth 1140, a spool 1200 disposed at the case 1100 and configured for a lace to be wound therearound, a knob 1400 covering the case 1100, and an engaging unit 1300 disposed at the case 1100 and coupled to the spool 1200. The engaging unit 1300 includes at least one pawl arm 1340 selectively coupled to at least one of the engaging teeth 1140, and at least one abutting member 1350. As the knob 1400 is rotated in a tensioning direction A2, the engaging unit 1300 is rotated and the at least one pawl arm 1340 is disengaged from at least one of the engaging teeth 1140 to allow the spool 1200 to tension the lace. As the knob 1400 is rotated in a releasing direction A1 to allow the at least one pawl arm 1340 to disengage from at least one of the engaging teeth 1140, the at least one abutting member 1350 is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.

[0059] Therefore, as the knob 1400 is rotated counterclockwise, the pawl arm 1340 is disengaged from the engaging teeth 1140, and since the abutting member 1350 can provide the friction, the lace will not be fully released and a tension thereof may be remained. The tension can then overcome the friction, and as the friction is overcome, the abutting member 1350 may be deflected to offset the friction temporarily, thereby releasing a small segment of the lace. The details of the fastening device 1000 will be described hereinafter.

[0060] The case 1100 may include a base 1110, a housing 1120, a partition 1130, a central hole and a lower ring groove 1150. The housing 1120 surrounds a receiving space and is detachably disposed at the base 1110. The partition 1130 extends radially and inward from the housing 1120 to separate the receiving space into an upper chamber S1 and a lower chamber S2. The central hole

penetrates the partition **1130** and the upper chamber **S1** is communicated with the lower chamber **S2**. The lower ring groove **1150** is located at the housing **1120** and is located at the lower chamber **S2**.

[0061] The spool **1200** includes an upper ring portion **1210**, a lower ring portion **1220** and a plurality of spool teeth **1230**. A winding track formed between the upper ring portion **1210** and the lower ring portion **1220** is configured for the lace to be wound therearound. The spool teeth **1230** are located at the upper ring portion **1210**. The spool **1200** may be received in the lower chamber **S2**, the lower ring portion **1220** may be correspondingly received in the lower ring groove **1150**, and the spool teeth **1230** may protrude into the upper chamber **S1** via the central hole.

[0062] The engaging unit **1300** may further include an upper disc **1310**, a lower disc **1320** and at least one driving arm **1330**, the upper disc **1310** and the lower disc **1320** are rotatably connected, the at least one pawl arm **1340** and the at least one abutting member **1350** are located at the lower disc **1320**, and the at least one driving arm **1330** is located at the upper disc **1310** and coupled to the knob **1400**. To be more specific, each of the upper disc **1310** and the lower disc **1320** is a circular-disc structure, and is received in the upper chamber **S1**. The upper disc **1310** may include an upper disc body and three coupling portions **1311**, the three coupling portions **1311** protrude downward and are arranged with intervals. The lower disc **1320** may include a lower disc body, three first radial tabs **1321** and three second radial tabs **1322**. The three first radial tabs **1321** and the three second radial tabs **1322** are disposed at the lower disc body alternatively. A coupling space is formed between each of the first radial tabs **1321** and each of the second radial tabs **1322** and is configured for each of the coupling portions **1311** to couple therewith; therefore, the upper disc **1310** and the lower disc **1320** may be rotatably connected. The engaging unit **1300** may further include a driving portion **1360** located at the lower disc **1320**, and the driving portion **1360** may correspondingly engage with the spool teeth **1230** extending into the upper chamber **S1**.

[0063] A number of the at least one pawl arm **1340** and a number of the at least one abutting member **1350** are three. Each pawl arm **1340** extends along the releasing direction **A1** from each second radial tab **1322**, and each abutting member **1350** extends along the releasing direction **A1** from each first radial tab **1321**. The pawl arm **1340** may include an engaging portion **1341** and a lower projection **1342**, the lower projection **1342** protrudes integrally and extends outward from the engaging portion **1341**, and the lower projection **1342** corresponds to at least one of the engaging teeth **1140**. Precisely, the engaging portion **1341** of the pawl arm **1340** may include a first engaging surface **1341a** (labeled in FIG. 4) and a second engaging surface **1341b** (labeled in FIG. 4). The first engaging surface **1341a** may correspond to the knob teeth **1410** and the engaging teeth **1140**. The lower projection **1342** extends outward from the second engaging surface **1341b** and corresponds to the engaging teeth **1140**, and the rest portion of the second engaging surface **1341b** may correspond to the knob teeth **1410**. The abutting member **1350** may include a pawl structure and is correspondingly engaged with at least one of the engaging teeth **1140**.

[0064] FIG. 4 is a cross-section top view of the fastening device **1000** of the first embodiment of FIG. 1. FIG. 5 is an operation shown by partial cross-section top views of the fastening device **1000** of the first embodiment of FIG. 1. The knob **1400** may include at least two knob teeth **1410**, one of the at least two knob teeth **1410** corresponds to the at least one driving arm **1330**, and another one of the at least two knob teeth **1410** corresponds to the pawl arm **1340**. As shown in FIGS. 2 to 4, a number of the at least one driving arm **1330** is three, a number of the at least two knob teeth **1410** is plural, the knob teeth **1410** are circularly arranged around an inner side wall of the knob **1400**, each of the knob teeth **1410** includes a first inclined surface **1411** and a second inclined surface **1412**, a slope of each of the first inclined surfaces **1411** is different from a slope of each of the second inclined surfaces **1412**, and an interval is contain between two of the knob teeth **1410** that are adjacent to each other. As the knob **1400** is rotated in the tensioning direction **A2**, the knob teeth **1410** push the driving arms **1330** to drive the engaging unit **1300** to rotate the spool **1200** in the tensioning direction **A2**, and as the knob **1400** is rotated in the releasing direction **A1**,

the knob teeth **1410** push and allow the pawl arms **1340** to disengage from at least one of the engaging teeth **1140**.

[0065] As shown in FIG. 4, each driving arm **1330** includes a pushed surface **1331**, and the pushed surface **1331** corresponds to the second inclined surface **1412** of one of the knob teeth **1410**. The engaging portion **1341** of each pawl arm **1340** may correspond to the interval between two adjacent knob teeth **1410** and is engaged with the engaging teeth **1140**. The abutting member **1350** is also engaged with the engaging teeth **1140**. At this time, if the user rotates the knob **1400** in the tensioning direction A2, the second inclined surfaces **1412** of the knob teeth **1410** push the pushed surfaces **1331** of the driving arms **1330**, thereby rotating the upper disc **1310**. Since the lower disc **1320** is rotatably connected to the upper disc **1310**, the lower disc **1320** will be affected thereby and is rotated in the tensioning direction A2. The lower projection **1342** that is in the same side with the second engaging surface **1341b** of the pawl arm **1340** is pushed by the engaging teeth **1140**, the pawl arm **1340** is deflected inward, and the abutting member **1350** is also deflected owing to the push of the engaging teeth **1140**, thereby allowing the spool **1200** to be rotated by the lower disc **1320** in the tensioning direction A2 for tensioning the lace. As the user stops forcing, the tension of the lace will pull the spool **1200** in the releasing direction A1, and because the pawl arm **1340** is engaged with the engaging teeth **1140**, rotation of the spool **1200** is prohibited, and the lace is fastened.

[0066] On the contrary, if the user rotates the knob **1400** in the releasing direction A1, the first inclined surfaces **1411** of the knob teeth **1410** push the second engaging surfaces **1341b** of the engaging portion **1341**. As shown in FIG. 5, the pawl arm **1340** is incrementally disengaged from the engaging teeth **1140** while the abutting member **1350** is still engaged with the engaging teeth **1140**. Because the abutting member **1350** is still engaged with the engaging teeth **1140**, a friction can be provided. Consequently, the lace will not be fully released at this time, and the tension of the lace that pulls the spool **1200** in the releasing direction A1 can be remained. As the tension of the lace is larger than the friction between the abutting member **1350** and the engaging teeth **1140**, the lace pulls the spool **1200** to rotate in the releasing direction A1. Since the pawl arm **1340** is separated from the engaging teeth **1140** owing to press of the knob teeth **1410**, the pawl arm **1340** cannot prohibit the spool **1200** from rotating in the releasing direction A1. Hence, the spool **1200** may be rotated by the tension of the lace to rotate the upper disc **1310** and the lower disc **1320** in the releasing direction A1. Meanwhile, the pawl arm **1340** will be disengaged from the knob teeth **1410** and is engaged with the engaging teeth **1140** again, which causes the spool **1200** to stop rotating, and the abutting member **1350** can also return to the status of engaging with the engaging teeth **1140**. Therefore, one segment of the lace can be released, and the goal of incremental releasing of the lace can be achieved.

[0067] FIG. 6 is one cross-section side view of the fastening device **1000** of the first embodiment of FIG. 1. FIG. 7 is another cross-section side view of the fastening device **1000** of the first embodiment of FIG. 1. The fastening device **1000** may include a connecting unit **1500**, and the connecting unit **1500** includes a central shaft **1510** and a screw member **1520**. The knob **1400** may further include a connecting boss **1420**, and the spool **1200** may further include a stopping portion **1250** and a positioning portion **1240**. The stopping portion **1250** extends from an inner edge of the upper ring portion **1210** toward an axial bore of the spool **1200**, and the positioning portion **1240** extends downward from the stopping portion **1250**. The central shaft **1510** may pass through the axial bore of the spool **1200** and pass the central hole of the case **1100** to pass through the engaging unit **1300**, and then may enter the connecting boss **1420**. The screw member **1520** may pass downward from the knob **1400** to insert and to screw with the central shaft **1510**. Therefore, the spool **1200**, the engaging unit **1300**, the knob **1400** and the housing **1120** may be integrally connected. In addition, the central shaft **1510** may include a positioning protrusion **1511**, and the positioning protrusion **1511** may be restricted by the positioning portion **1240**.

[0068] As shown in FIG. 6, the knob **1400** is not pulled up and is in the first position, at this time

the positioning protrusion **1511** is located below the positioning portion **1240**, and the fastening device **1000** can be operated to tension or incrementally release the lace. On the contrary, as shown in FIG. 7, as fully releasing the lace, the knob **1400** may be pulled upward to the second position. [0069] The central shaft **1510** will move upward to allow the positioning protrusion **1511** to be located above the positioning portion **1240**, and the positioning protrusion **1511** will be restricted by the stopping portion **1250** of the spool **1200** and cannot separate from the spool **1200**. Since the engaging unit **1300** is abutted against a shoulder of the central shaft **1510**, the engaging unit **1300** will be moved upward by the central shaft **1510**. The spool teeth **1230** will be disengaged from the driving portion **1360**, and free rotation of the spool **1200** will not be restricted by the engaging unit **1300**, thereby fully releasing the lace.

[0070] FIG. 8 is a three-dimensional schematic view of a fastening device **2000** according to a second embodiment of the present disclosure. FIG. 9 is one exploded view of the fastening device **2000** of the second embodiment of FIG. 8. FIG. 10 is another exploded view of the fastening device **2000** of the second embodiment of FIG. 8. FIG. 11 is a cross-section top view of the fastening device **2000** of the second embodiment of FIG. 8. The fastening device **2000** of the second embodiment is similar to the fastening device **1000** of the first embodiment and includes a case (not labeled in the second embodiment), a spool **2200**, an engaging unit **2300**, a knob **2400** and a connecting unit (not labeled in the second embodiment). The case includes a housing **2120** and a plurality of engaging teeth **2140**. The engaging unit **2300** includes an upper disc **2310**, a lower disc **2320**, driving arms **2330**, pawl arms **2340** and abutting members **2350**. The knob **2400** includes a plurality of knob teeth **2410**, and only the differences thereof are mentioned hereinafter. It is noted that, in the second embodiment, the base of the case is omitted.

[0071] The knob teeth **2410** are located at an inner side wall of the knob **2400**. The engaging portion of the pawl arm **2340** does not include any lower projection. The upper disc **2310** includes three engaging cavities **2312**, the lower disc **2320** includes three radial tabs **2323**, and each of the radial tabs **2323** is correspondingly engaged with each of the engaging cavities **2312**. Each of the pawl arms **2340** extends from one side of each radial tab **2323** toward the releasing direction **A1**, and each of the abutting members **2350** extends from another side of each radial tab **2323** toward the tensioning direction **A2**.

[0072] FIG. 12 is one cross-section side view of the fastening device **2000** of the second embodiment of FIG. 8. FIG. 13 is another cross-section side view of the fastening device **2000** of the second embodiment of FIG. 8. As shown in FIG. 12, the knob **2400** is not pulled up and is in the first position, rotating the knob **2400** in the tensioning direction **A2** may allow the spool **2200** to tension the lace, and rotating the knob **2400** in the releasing direction **A1** may deflect the pawl arm **2340** to incrementally release the lace. On the contrary, as shown in FIG. 13, the knob **2400** may be pulled upward to the second position, and the lace may be fully released.

[0073] FIG. 14 is one exploded view of a fastening device **3000** according to a third embodiment of the present disclosure. FIG. 15 is another exploded view of the fastening device **3000** of the third embodiment of FIG. 14. The fastening device **3000** of the third embodiment is similar to the fastening device **1000** of the first embodiment and includes a case (not labeled in the third embodiment), a spool **3200**, an engaging unit **3300**, a knob **3400** and a connecting unit (not labeled in the third embodiment). The case includes a housing **3120** and a plurality of engaging teeth **3140**. The engaging unit **3300** includes an upper disc **3310**, a lower disc **3320**, driving arms **3330**, pawl arms **3340** and abutting members **3350**. The knob **3400** includes a plurality of knob teeth **3410**, and only the differences thereof are mentioned hereinafter. It is noted that, in the third embodiment, the base of the case is omitted.

[0074] The knob teeth **3410** are located at an inner side wall of the knob **3400**, and each of the knob teeth **3410** includes a first inclined surface **3411** (labeled in FIG. 17), a second inclined surface **3412** (labeled in FIG. 17) and a third inclined surface **3413** (labeled in FIG. 17). The third inclined surface **3413** is connected between the first inclined surface **3411** and the second inclined

surface **3412**, and no interval is contained between each knob tooth **3410** as shown in the first embodiment.

[0075] FIG. **16** is a three-dimensional cross-section view of the fastening device **3000** of the third embodiment of FIG. **14**. FIG. **17** is a cross-section top view of the fastening device **3000** of the third embodiment of FIG. **14**. Please refer to FIGS. **16** to **17** with references of FIGS. **14** to **15**, the upper disc **3310** includes three engaging cavities **3312**, the lower disc **3320** includes three radial tabs **3323**, and each of the radial tabs **3323** is correspondingly engaged with each of the engaging cavities **3312**. Each of the pawl arms **3340** may include a lower arm portion **3343** and an upper protrusion **3344**, the lower arm portion **3343** extends from one side of each radial tab **3323** toward the releasing direction **A1**, and a distal end of the lower arm portion **3343** may be engaged with the engaging teeth **3140**. The upper protrusion **3344** is located on the top of the distal end of the lower arm portion **3343** and is correspondingly engaged with the knob teeth **3410**.

[0076] The abutting member **3350** may be located below the pawl arm **3340** and has a strip structure. The case may include a plurality of stopping curved ribs **3160** protruding upward from the partition (not labeled in the third embodiment) and corresponding to the abutting member **3350**. The stopping curved ribs **3160** may be circularly arranged on the partition. More particularly, the extending direction of the abutting member **3350** is perpendicular to the extending direction of the lower arm portion **3343**. The extending direction of the abutting member **3350** is not parallel to the stopping curved rib **3160**, and the abutting member **3350** is in a gap formed between two adjacent stopping curved ribs **3160** in a general status.

[0077] FIG. **18** is one cross-section side view of the fastening device **3000** of the third embodiment of FIG. **14**. FIG. **19** is another cross-section side view of the fastening device **3000** of the third embodiment of FIG. **14**. As shown in FIG. **18**, the knob **3400** is located in a first position. As the knob **3400** is rotated in the tensioning direction **A2** to rotate the engaging unit **3300**, the pawl arm **3340** will be disengaged from the engaging teeth **3140**, and the abutting member **3350** will be disengaged from the gap between two adjacent stopping curved ribs **3160**, thereby allowing the spool **3200** to tension the lace. On the contrary, as the knob **3400** is rotated in the releasing direction **A1**, the knob teeth **3410** deflect the upper protrusion **3344** while the abutting member **3350** is still located at the gap between two adjacent stopping curved ribs **3160**, the lace will not be fully released at this time, and the tension of the lace to pull the spool **3200** in the releasing direction **A1** is remained. As the tension of the lace is larger than a friction between the abutting member **3350** and the stopping curved ribs **3160**, the tension of the lace pulls the spool **3200** to rotate in the releasing direction **A1**, the abutting member **3350** is deflected, and the spool **3200** may be rotated in the releasing direction **A1**. Then, the pawl arm **3340** is engaged with the engaging teeth **3140** again, which causes the spool **3200** to stop rotating, and the abutting member **3350** can also return to the gap between two adjacent stopping curved ribs **3160**. Therefore, one segment of the lace can be released, and the goal of incremental releasing of the lace can be achieved. Moreover, as shown in FIG. **19**, as the knob **3400** is pulled upward to be located at a second position, the lace may be fully released.

[0078] FIG. **20** is one exploded view of a fastening device **4000** according to a fourth embodiment of the present disclosure. FIG. **21** is another exploded view of the fastening device **4000** of the fourth embodiment of FIG. **20**. The fastening device **4000** includes a case (not labeled in the fourth embodiment) including a plurality of engaging teeth **4140**, a spool **4200** disposed at the case and configured for a lace to be wound therearound, a knob **4400** covering the case and including a plurality of knob teeth **4410**, and an engaging unit **4300** disposed at the case and coupled to the spool **4200**. The engaging unit **4300** includes at least one pawl arm **4340** selectively coupled to at least one of the engaging teeth **4140**. As the knob **4400** is rotated in a tensioning direction **A2**, the engaging unit **4300** is rotated to drive the spool **4200** to tension the lace, and as the knob **4400** is rotated in a releasing direction **A1**, the at least one pawl arm **4340** is incrementally and radially deflected, a tension of the lace causes the at least one pawl arm **4340** to be fully disengaged from at

least one of the engaging teeth **4140** to offset a friction temporarily, thereby allowing the lace to be incrementally released. It is noted that, the base of the case is omitted in the fourth embodiment, and the structure similar to the first embodiment will not be repeated.

[0079] Precisely, the case may further include a housing **4120** for the engaging teeth **4140** to be disposed thereat. The knob teeth **4410** are located at an inner side wall of the knob **4400**, and no interval shown in the first embodiment is contained between the knob teeth **4410** that are adjacent to each other. The shape of the knob tooth **4410** is identical to the shape of the engaging tooth **4140**.

[0080] The engaging unit **4300** may further include an upper disc **4310**, a lower disc **4320** and driving arms **4330**, the lower disc **4320** is located below the upper disc **4310**, the pawl arms **4340** are located at the lower disc **4320**, the driving arms **4330** are located at the upper disc **4310**, and the driving arms **4330** are coupled to the knob teeth **4410**.

[0081] FIG. 22 is one cross-section top view of the fastening device **4000** of the fourth embodiment of FIG. 20. FIG. 23 is another cross-section top view of the fastening device **4000** of the fourth embodiment of FIG. 20. FIG. 24 is one cross-section side view of the fastening device **4000** of the fourth embodiment of FIG. 20. FIG. 25 is another cross-section side view of the fastening device **4000** of the fourth embodiment of FIG. 20. As shown in FIGS. 20 to 22, the upper disc **4310** may include three accommodating grooves **4313** arranged with intervals. The lower disc **4320** may include three radial tabs **4323**, each of the radial tabs **4323** may be received in each of the accommodating grooves **4313**, and the upper disc **4310** is allowed to be rotated relative to the lower disc **4320** in a range within each of the accommodating groove **4313**. In addition, the engaging unit **4300** may further include three pressing blocks **4360**, three rotating arms **4370** and three pushing blocks **4380**. The three rotating arms **4370** are disposed at the upper disc **4310** with intervals and correspond to the knob teeth **4410**. The three pushing blocks **4380** are also disposed at the upper disc **4310** with intervals and extend downward. The three pressing blocks **4360** are disposed at the lower disc **4320** with intervals and respectively correspond to the three rotating arms **4370**. As the knob **4400** is rotated in the releasing direction **A1**, the rotating arm **4370** is pressed by the pressing block **4360**, and the upper disc **4310** is allowed to rotate relative to the lower disc **4320**, thereby allowing the pushing block **4380** to incrementally deflect the pawl arm **4340**. In other embodiments, a number of the pressing blocks, a number of the rotating arms, a number of the pushing blocks and a number of the pawl arms are at least one, and the present disclosure is not limited thereto.

[0082] Hence, as shown in FIGS. 22 and 24, the knob **4400** is not pulled upward and is in the first position. As the knob **4400** is rotated in the tensioning direction **A2** to rotate the upper disc **4310** for allowing one side wall of the accommodating groove **4313** to push the radial tab **4323**, the lower disc **4320** will also be rotated to drive the spool **4200** to tension the lace. As the knob **4400** stops rotating, the lace cannot be released owing to the engagement between the pawl arm **4340** and the engaging teeth **4140**.

[0083] On the contrary, as the knob **4400** is rotated in the releasing direction **A1**, the knob teeth **4410** push the rotating arms **4370** in the releasing direction **A1**, and the rotating arms **4370** are pressed by the pressing blocks **4360** so as to disengage from the knob teeth **4410**, thereby allowing the upper disc **4310** to rotate relative to the lower disc **4320**. Meanwhile, the pushing blocks **4380** close the pawl arms **4340** to incrementally deflect the pawl arms **4340**, but since the pawl arms **4340** are not fully disengaged from the engaging teeth **4140**, the lace will not be fully released at this time, and the tension of the lace that pulls the spool **4200** in the releasing direction **A1** can be remained. As the tension of the lace is larger than the friction between the pawl arms **4340** and the engaging teeth **4140**, the pawl arms **4340** will be fully released to allow the lace to pull the spool **4200** in the releasing direction **A1** such that the pushing blocks **4380** deflect the pawl arms **4340** to allow the pawl arms **4340** to engage with the engaging teeth **4140** again, and the spool **4200** stops rotating. Therefore, one segment of the lace can be released, and the goal of incremental releasing

of the lace can be achieved. It is noted that, the pawl arm **4340** may include a notch to assist the pawl arm **4340** to be smoothly released, but the present disclosure is not limited thereto. Furthermore, as shown in FIG. **25**, the knob **4400** may be pulled up to a second position, and the lace may be fully released.

[0084] FIG. **26** is one exploded view of a fastening device **5000** according to a fifth embodiment of the present disclosure. FIG. **27** is another exploded view of the fastening device **5000** of the fifth embodiment of FIG. **26**. The fastening device **5000** of the fifth embodiment is similar to the fastening device **1000** of the first embodiment and includes a case (not labeled in the fifth embodiment), a spool **5200**, an engaging unit **5300**, a knob **5400** and a connecting unit (not labeled in the fifth embodiment). The case includes a housing **5120** and a plurality of engaging teeth **5140**. The engaging unit **5300** includes an upper disc **5310**, a lower disc **5320**, driving arms **5330**, pawl arms **5340** and abutting members **5350**. The knob **5400** includes a plurality of knob teeth **5410**, and only the differences thereof are mentioned hereinafter. It is noted that, in the fifth embodiment, the base of the case is omitted.

[0085] The knob teeth **5410** are located at an inner side wall of the knob **5400**. The upper disc **5310** includes three engaging cavities **5312**, the lower disc **5320** includes three radial tabs **5323**, and each of the radial tabs **5323** is correspondingly engaged with each of the engaging cavities **5312**. Each of the pawl arms **5340** extends from one side of each radial tab **5323** toward the releasing direction **A1**, and each of the abutting members **5350** may have a toothed structure and is located at an inner side of each of the pawl arms **5340**.

[0086] The case may further include a gear ring **5170** and a plurality of stopping teeth **5180**. The gear ring **5170** extends upward from the partition **5130**, and the stopping teeth **5180** are located at the gear ring **5170** and face toward the engaging teeth **5140**.

[0087] FIG. **28** is one cross-section top view of the fastening device **5000** of the fifth embodiment of FIG. **26**. FIG. **29** is another cross-section top view of the fastening device **5000** of the fifth embodiment of FIG. **26**. FIG. **30** is one cross-section side view of the fastening device **5000** of the fifth embodiment of FIG. **26**. FIG. **31** is another cross-section side view of the fastening device **5000** of the fifth embodiment of FIG. **26**. As shown in FIGS. **28** and **30**, the knob **5400** is not pulled up and is in the first position, the lower disc **5320** may be located above the partition **5130**, the pawl arm **5340** corresponds to the engaging teeth **5140**, and the abutting member **5350** faces the gear ring **5170** but not engages with the stopping teeth **5180**. As rotating the knob **5400** in the tensioning direction **A2** to rotate the engaging unit **5300**, the pawl arm **5340** will be disengaged from the engaging teeth **5140**, thereby allowing the spool **5200** to tension the lace.

[0088] On the contrary, as shown in FIG. **29**, as the knob **5400** is rotated in the releasing direction **A1**, the knob teeth **5410** deflect the pawl arm **5340** to allow the pawl arm **5340** to disengage from the engaging teeth **5140**, the pawl arm **5340** swings inward such that the abutting member **5350** is engaged with the stopping teeth **5180**, the lace will not be fully released at this time, and the tension of the lace to pull the spool **5200** in the releasing direction **A1** is remained. As the tension of the lace is larger than a friction between the abutting member **5350** and the stopping teeth **5180**, the tension of the lace pulls the spool **5200** to rotate in the releasing direction **A1**, the abutting member **5350** is deflected, and the spool **5200** may be rotated in the releasing direction **A1**. Then, the pawl arm **5340** is engaged with the engaging teeth **5140** again, which causes the spool **5200** to stop rotating, and the abutting member **5350** is disengaged from the stopping teeth **5180**. Therefore, one segment of the lace can be released, and the goal of incremental releasing of the lace can be achieved. Moreover, as shown in FIG. **31**, as the knob **5400** is pulled upward to be located at a second position, the lace may be fully released.

[0089] FIG. **32** is one exploded view of a fastening device **6000** according to a sixth embodiment of the present disclosure. FIG. **33** is another exploded view of the fastening device **6000** of the sixth embodiment of FIG. **32**. FIG. **34** is a partial three-dimensional schematic view of the fastening device **6000** of the sixth embodiment of FIG. **32**. The fastening device **6000** of the sixth

embodiment is similar to the fastening device **1000** of the first embodiment and includes a case (not labeled in the sixth embodiment), a spool **6200**, an engaging unit **6300**, a knob **6400** and a connecting unit (not labeled in the sixth embodiment). The case includes a housing **6120** and a plurality of engaging teeth **6140**. The engaging unit **6300** includes an upper disc **6310**, a lower disc **6320**, driving arms **6330**, pawl arms **6340** and abutting members **6350**. The knob **6400** includes a plurality of knob teeth **6410**, and only the differences thereof are mentioned hereinafter. It is noted that, in the sixth embodiment, the base of the case and the screw member of connecting unit is omitted.

[0090] As shown in FIGS. **32** and **33**, the engaging unit **6300** may further include three resting blocks **6390**, each resting block **6390** is located between a distal end of the abutting member **6350** and a proximal end of the pawl arm **6340** adjacent thereto, and each resting block **6390** is configured to correspond to each driving arm **6330** at the upper disc **6310**. The upper disc **6310** may include three stopping blocks **6314** extending radially and outward therefrom. Each stopping block **6314** may be located between two adjacent driving arms **6330** and correspond to each pawl arm **6340** at the lower disc **6320**. The pawl arm **6340** may include a lower arm portion **6343** and an upper protrusion **6344**, the lower arm portion **6343** is configured to engage with the engaging teeth **6140**, and the upper protrusion **6344** extends upward and integrally from the lower arm portion **6343** along an axial direction. A radial distance between the outer most edge of the upper protrusion **6344** and a center of the lower disc **6320** is larger than a radial distance between the outer most edge of the lower arm portion **6343** and the center of the lower disc **6320**. Hence, as the upper disc **6310** is engaged with the lower disc **6320**, the upper disc **6310** is rotatable with the lower disc **6320**, and the upper protrusion **6344** will protrude into the gap between the driving arm **6330** and the stopping block **6314**. One side surface of the upper protrusion **6344** is abutted against one side surface of the stopping block **6314**, and one side surface of the driving arm **6330** is abutted against one side surface of the resting block **6390**.

[0091] The case may include a plurality of depressions **6190** located at the partition. In other words, with the axial concave and convex shapes on the upper surface of the partition, the depressions **6190** can be formed on the partition. The abutting member **6350** with a toothed structure can correspond to at least one of the depressions **6190**. The spool **6200** may include a positioning portion **6240** extends upward, and the positioning portion **6240** may be configured to position the positioning protrusion of the central shaft.

[0092] Moreover, as shown in FIG. **33**, the knob teeth **6410** of the knob **6400** are not circularly arranged at an inner side wall of the knob **6400**, and six knob teeth **6410** are arranged with intervals, but the present disclosure is not limited thereto.

[0093] FIG. **35** is one cross-section top view of the fastening device **6000** of the sixth embodiment of FIG. **32**. FIG. **36** is another cross-section top view of the fastening device **6000** of the sixth embodiment of FIG. **32**. FIG. **37** is one cross-section side view of the fastening device **6000** of the sixth embodiment of FIG. **32**. FIG. **38** is another cross-section side view of the fastening device **6000** of the sixth embodiment of FIG. **32**. As shown in FIGS. **34**, **35** and **37**, the knob **6400** is not pulled up and is in the first position, the positioning protrusion of the central shaft is located below the positioning portion **6240** (the positioning portion **6240** is not illustrated in the cross-section side view in the drawings owing to the view angle thereof), and the lower arm portion **6343** of the pawl arm **6340** is correspondingly engaged with the engaging teeth **6140**. As the knob **6400** is rotated in the tensioning direction **A2**, the knob teeth **6410** of the knob **6400** push the driving arms **6330** to rotate the lower disc **6320**, and the spool **6200** is rotated to tension the lace. As the knob **6400** stops rotating, the lace may not be released owing to the engagement between the pawl arm **6340** and the engaging teeth **6140**. It is noted that, since the driving arm **6330** will rest on the resting block **6390** as being forced, broken of the driving arm **6330** may be avoided, and the structural strength may be increased.

[0094] As shown in FIGS. **34**, **36** and **37**, as the knob **6400** is rotated in the releasing direction **A1**,

the upper protrusion **6344** of the pawl arm **6340** is deflected as being pushed by the knob teeth **6410**, and the lower arm portion **6343** of the pawl arm **6340** is disengaged from the knob teeth **6410**. Since the abutting member **6350** is engaged with the depressions **6190**, the lace will not be fully released at this time, and the tension of the lace to pull the spool **6200** in the releasing direction **A1** is remained. As the tension of the lace is larger than a friction between the abutting member **6350** and the depressions **6190**, the tension of the lace pulls the spool **6200** to rotate in the releasing direction **A1**, the abutting member **6350** is deflected, and the spool **6200** may be rotated in the releasing direction **A1**. Then, the lower arm portion **6343** of the pawl arm **6340** is engaged with the engaging teeth **6140** again, and the spool **6200** is allowed to stop rotating. Therefore, one segment of the lace can be released, and the goal of incremental releasing of the lace can be achieved. It is noted that, since the upper protrusion **6344** of the pawl arm **6340** will be restricted by the stopping block **6314**, broken of the pawl arm **6340** is avoided, and the structural strength is increased. Furthermore, it is also noted that, as the knob **6400** is rotated in the releasing direction **A1**, the knob teeth **6410** use a first force to deflect the pawl arm **6340** and use a second force to deflect the driving arm **6330**, and the first force and the second force are smaller than the friction between the abutting members **6350** and the depressions **6190**, thereby allowing the abutting members **6350** to remain the tension of the lace. In other embodiments, if the knob deflects the pawl arms and the driving arms at the same time, a sum of the first force and the second force is smaller than the friction generated from the abutting members.

[0095] Moreover, as shown in FIG. **38**, as the knob **6400** is pulled upward to be located at a second position, the positioning protrusion of the central post is above the positioning portion **6240**, and the lace may be fully released.

[0096] FIG. **39** is one exploded view of a fastening device **7000** according to a seventh embodiment of the present disclosure. FIG. **40** is another exploded view of the fastening device **7000** of the seventh embodiment of FIG. **39**. The fastening device **7000** of the seventh embodiment is similar to the fastening device **6000** of the six embodiment and includes a case (not labeled in the seventh embodiment), a spool **7200**, a knob **7400**, at least one pawl arm **7340** and at least one abutting member **7260**. The case includes a housing **7120**, a plurality of engaging teeth **7140** and a base **7110**. The knob **7400** includes a plurality of knob teeth **7410**. Similarly, as the knob **7400** is rotated in a releasing direction **A1** to allow the at least one pawl arm **7340** to disengage from at least one of the engaging teeth **7140**, the at least one abutting member **7260** is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.

[0097] In the seventh embodiment, the fastening device **7000** also includes an engaging unit **7300**, and the engaging unit **7300** includes an upper disc **7310** (labeled in FIG. **41**), a lower disc **7320** (labeled in FIG. **41**), at least one driving arm **7330** and the pawl arm **7340**. The difference is that the engaging unit **7300** does not include the abutting members **6350** located at the lower disc **6320** as the engaging unit **6300**, and, as shown in FIGS. **39** and **40**, the spool **7200** includes the abutting members **7260** extending downward from the lower ring portion instead. Moreover, a plurality of depressions **7190** of the case are located at a surface of the base **7110**.

[0098] FIG. **41** is one cross-section side view of the fastening device **7000** of the seventh embodiment of FIG. **39**. FIG. **42** is another cross-section side view of the fastening device **7000** of the seventh embodiment of FIG. **39**. Please refer to FIGS. **41** and **42** with references of FIGS. **39** and **40**, as the knob **7400** is rotated in the releasing direction **A1**, the knob teeth **7410** deflect the pawl arm **7340** to disengage from the engaging teeth **7140**. Since the abutting members **7260** are engaged with the depressions **7190**, the lace will not be fully released at this time, and the tension of the lace to pull the spool **7200** in the releasing direction **A1** is remained. As the tension of the lace is larger than a friction between the abutting members **7260** and the depressions **7190**, the tension of the lace pulls the spool **7200** to rotate in the releasing direction **A1**, the abutting member **7260** is deflected, and the spool **7200** may be rotated in the releasing direction **A1**. Then, the pawl

arm **7340** is engaged with the engaging teeth **7140** again, and the spool **7200** is allowed to stop rotating. Therefore, one segment of the lace can be released, and the goal of incremental releasing of the lace can be achieved. Moreover, as shown in FIG. **42**, as the knob **7400** is pulled upward to be located at a second position, the lace may be fully released.

[0099] FIG. **43** is one exploded view of a fastening device **8000** according to an eighth embodiment of the present disclosure. FIG. **44** is another exploded view of the fastening device **8000** of the eighth embodiment of FIG. **43**. FIG. **45** is one cross-section side view of the fastening device **8000** of the eighth embodiment of FIG. **43**. FIG. **46** is another cross-section side view of the fastening device **8000** of the eighth embodiment of FIG. **43**. The fastening device **8000** of the eighth embodiment is similar to the fastening device **7000** of the seventh embodiment and includes a case (not labeled in the eighth embodiment), a spool **8200**, a knob **8400** and an engaging unit **8300**. The case includes a housing **8120**, a plurality of engaging teeth **8140** and a plurality of depressions **8190**. The difference is that the depressions **8190** are located below the partition, and abutting members **8260** protrude upward from the upper ring portion of the spool **8200**. Other similar structures and motions will not be repeated.

[0100] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0101] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure covers modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A fastening device, comprising: a case comprising a plurality of engaging teeth; a spool disposed at the case and configured for a lace to be wound therearound; a knob covering the case; at least one pawl arm selectively coupled to at least one of the engaging teeth; and at least one abutting member; wherein as the knob is rotated in a tensioning direction, the at least one pawl arm is disengaged from at least one of the engaging teeth to allow the spool to tension the lace, and as the knob is rotated in a releasing direction to allow the at least one pawl arm to disengage from at least one of the engaging teeth, the at least one abutting member is deflected by a tension of the lace to offset a friction temporarily, thereby allowing the lace to be incrementally released.
2. The fastening device of claim 1, further comprising an engaging unit disposed at the case and coupled to the spool, wherein the engaging unit comprises an upper disc, a lower disc and at least one driving arm, the upper disc and the lower disc are rotatably connected, the at least one pawl arm and the at least one abutting member are located at the lower disc, and the at least one driving arm is located at the upper disc and coupled to the knob.
3. The fastening device of claim 2, wherein the knob comprises at least two knob teeth, one of the at least two knob teeth corresponds to the at least one driving arm, another one of the at least two knob teeth corresponds to the at least one pawl arm, as the knob is rotated in the tensioning direction, the one of the at least two knob teeth pushes the at least one driving arm to drive the engaging unit to rotate the spool in the tensioning direction, and as the knob is rotated in the releasing direction, the another one of the at least two knob teeth pushes and allows the at least one pawl arm to disengage from at least one of the engaging teeth.
4. The fastening device of claim 3, wherein the at least one pawl arm comprises: an engaging portion; and a lower projection extending from the engaging portion integrally and outward, the lower projection corresponding to at least one of the engaging teeth.
5. The fastening device of claim 3, wherein a number of the at least two knob teeth is plural, the knob teeth are circularly arranged around an inner side wall of the knob, each of the knob teeth

comprises a first inclined surface and a second inclined surface, a slope of each of the first inclined surfaces is different from a slope of each of the second inclined surfaces, and an interval is contained between two of the knob teeth that are adjacent thereto.

6. The fastening device of claim 2, wherein the case further comprises a housing and a partition, the partition extends inward from the housing and an upper chamber and a lower chamber are formed, and the engaging unit is received in the upper chamber.

7. The fastening device of claim 6, wherein the at least one abutting member is located below the at least one pawl arm and has a strip structure, and the case comprises a plurality of stopping curved ribs protruding upward from the partition and corresponding to the at least one abutting member.

8. The fastening device of claim 6, wherein the at least one abutting member has a toothed structure and is located at an inner side of the at least one pawl arm, the case further comprises a gear ring and a plurality of stopping teeth, the gear ring extends upward from the partition, the stopping teeth are located at the gear ring and face toward the engaging teeth, and as the knob is rotated in the releasing direction and the at least one pawl arm is deflected to disengage from at least one of the engaging teeth, the at least one abutting member is engaged with at least one of the stopping teeth.

9. The fastening device of claim 6, wherein the case further comprises a plurality of depressions, the depressions are located at the partition, and the at least one abutting member corresponds to at least one of the depressions.

10. The fastening device of claim 1, wherein the at least one abutting member has a pawl structure and is correspondingly engaged with at least one of the engaging teeth.
